

User Experience and Design

Computing and Mathematics

May, 2026



User Experience and Design (A13384)

Short Title:	User Experience and Design	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Graphic Design and Animation	
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Credits	10	Level: Postgraduate

Description of Module / Aims

User Experience (UX) is a concept that has many dimensions, and it includes a variety of different disciplines? such as interaction design, information architecture, visual design, usability, and human-computer interaction. The aim of this module is for the students to learn, through information, demonstration, exploration and application, a human-centred approach to design and evaluation of a wide range of interactive systems and products, and their contexts of use.

Programmes

COMP-0619	MSc in Computer Science (Enterprise Software Systems) (WD_KCESS_R)	1 / 0 / E
COMP-0619	MSc in Computing (Information Systems Processes) (WD_KISYP_R)	1 / 0 / E

Indicative Content

- Visual & Interaction Design
- Information Architecture
- Usability & Accessibility
- Human Computer Interaction (HCI) & User Centred Design (UCD)
- Ergonomics
- Evaluation & Testing

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the theoretical relevance, and the application, of human physical, cognitive, social and affective knowledge, to the design of interactive systems.
2. Critique and evaluate the user requirements for an interactive system or product.
3. Design, implement and evaluate interactive computing systems.
4. Interpret outcomes of study and research in writing and orally.

Learning and Teaching Methods

- Lectures -- will be used to introduce and discuss UX concepts.
- Practicals/Workshops -- will be used to develop and apply the concepts introduced and discussed during lectures.
- Student Independent Learning -- Students will research and apply the various concepts introduced and discussed during lectures.
- Students will use Microsoft Office and Adobe Creative Cloud.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4
<i>Assignment</i>	20%	4
<i>Assignment</i>	80%	1,2,3,4

Assessment Criteria

- <40%:** Unable to interpret and describe key concepts of the specific knowledge domain(s).
- 40%-59%:** Be able to interpret and describe key concepts of the specific knowledge domain(s). Be able to discuss key concepts of the specific knowledge domain and ability to discover and integrate related knowledge in other knowledge domains.
- 60%-69%:** Be able to solve problems within the specific knowledge domain(s) by experimenting with the appropriate skills and tools.
- 70%-100%:** All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	24
Practical	24	24
Independent Learning	222	222

Essential Material

- "Fundamental Principles of Great UX Design: How to Deliver Great UX Design." <http://www.uxmatters.com/mt/archives/2014/11/fundamental-principles-of-great-ux-design-how-to-deliver-great-ux-design.php/#sthash.a4jgbhYc.dpuf>
- "Interaction Design." <http://www.id-book.com/index.php>
- Preece, J. and Rogers, Y and Sharp, H. _Interaction Design: Beyond Human Computer Interaction_. 4th Edition. Chichester, UK. : Wiley., 2015.

Supplementary Material

- "Color in Motion." <http://www.mariaclaudiacortes.com/colors/Colors.html>
- "UX Books." <http://uxdesign.cc/ux-books/>
- Benyon, D. and Turner, P. and Turner, S. _Designing Interactive Systems: People, Activities, Contexts, Technologies_. Essex, UK: Addison-Wesley, 2005.
- Garrett, J. _The Elements of User Experience: User Centred Design for the Web_. New York: New Riders, 2010.
- Heim, S. _The Resonant Interface: HCI Foundations for Interaction Design_. Boston, MA: Addison-Wesley, 2008.
- Johnson, J. _Designing with the Mind in Mind_. 2nd Edition. Waltham, MA: Morgan Kaufman, 2014.
- McManus, S. _Web Design: make your website a success_. 6th Edition. Warwickshire, England: In Easy Steps, 2014.

- Raskin, J. _The Humane Interface: New Directions for Designing__Interactive Systems_. Boston, MA: Addison-Wesley, 2000.

Requested Resources

- Room Type: Computer Lab
- Lecture Room: Loose Seated